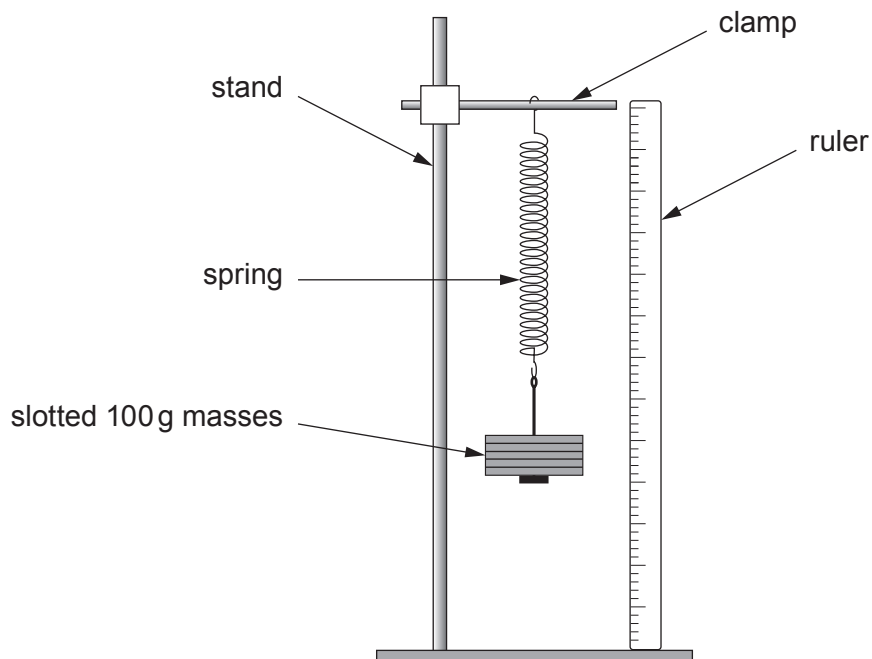


Examiner
only

4. (a) Describe how the apparatus shown below is used to investigate the extension of a spring. [6 QER]



Include the following in your answer:

- how the apparatus is set up,
- a description of the method used to obtain the results,
- how the results are analysed.



(b) A spring is 3 cm long. When an object is added to it, its length increases to 18 cm.

(i) Calculate the extension produced by the object. [1]

extension = cm

(ii) The spring constant of the spring is 0.8 N/cm.
Use an equation from page 2 to calculate the force produced by the object on the spring. [2]

force = N

(iii) This force is the weight of the object.
Use the equation:

$$\text{mass} = \frac{\text{weight}}{\text{gravitational field strength}}$$

to calculate the mass of the object. (Gravitational field strength, $g = 10 \text{ N/kg}$) [2]

mass = kg

